

Coding Theory: Introduction

Lecture Note Version in LaTeX

These notes reorganize an introductory slide deck into a self-contained lecture note. The goal is not only to transcribe the slides, but also to make the ideas readable as a continuous mathematical document.

Topic	Introductory coding theory
Focus	Reliability of transmission and storage
Main examples	Erasures, errors, discrete/continuous channels, storage, BEC
Document type	Clean article-style notes with TikZ diagrams

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1 What is Coding Theory?

Basic idea

Coding theory studies how to design mechanisms that allow us to recover or protect data when the data is disturbed during transmission or storage.

At the most basic level, the problem is simple to state:

original data $\longrightarrow$ perturbation or noise $\longrightarrow$ corrupted data.

The mathematical goal is to infer the original information from what is received.

Core goal

Given corrupted data, recover the intended data as reliably as possible.

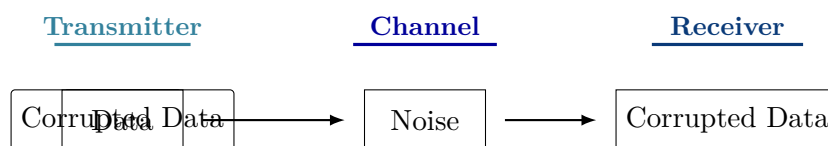


This introductory picture already contains the main philosophy of the subject: we do not usually control the channel or storage medium perfectly, so instead we design redundancy, structure, and decoding rules that make recovery possible.

2 Communication and Storage Models

Coding theory is usually explained through a channel model. A sender transmits data, a channel perturbs it, and a receiver observes an altered version.

2.1 Communication channel model



A communication system can be abstracted as

transmitter $\longrightarrow$ channel $\longrightarrow$ receiver.

The channel introduces uncertainty, commonly modeled as noise.

2.2 Erasure channels

An *erasure* means that some positions are lost, but the receiver knows exactly which positions are missing.

$$(x_1, x_2, x_3, x_4, x_5) \longrightarrow \text{erasures} \longrightarrow (x_1, *, x_3, *, *).$$

This is easier than arbitrary error correction, because the receiver is told where the uncertainty occurs.

Erasures vs. errors

In an erasure channel, the bad locations are visible. In an error channel, the bad locations are typically invisible.

2.3 Channels with errors

In an error channel, transmitted symbols may be changed:

$$(x_1, x_2, x_3, x_4, x_5) \longrightarrow \text{errors} \longrightarrow (y_1, y_2, y_3, y_4, y_5).$$

Usually we do *not* know where or when the errors occurred. That is why error correction is subtler than erasure correction.

2.4 Discrete channels with errors

For a discrete channel, the symbols belong to a finite alphabet A :

$$(x_1, x_2, x_3, x_4, x_5) \in A^5, \quad |A| < \infty.$$

The received word also lies in A^5 , but some coordinates may have changed. This is the classical setting for block codes over finite alphabets.

2.5 Continuous channels with errors

In a continuous channel, transmitted vectors live in a continuous space such as $\mathbb{R}^5$ or $\mathbb{C}^5$:

$$(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \text{ or } \mathbb{C}^5.$$

This model is closer to many physical communication systems, where signals are analog or are represented by real/complex amplitudes.

2.6 Multi-input multi-output channels

A MIMO channel can be modeled by a matrix input and matrix output, for example

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \in \mathbb{C}^{2 \times 2}, \quad Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \in \mathbb{C}^{2 \times 2}.$$

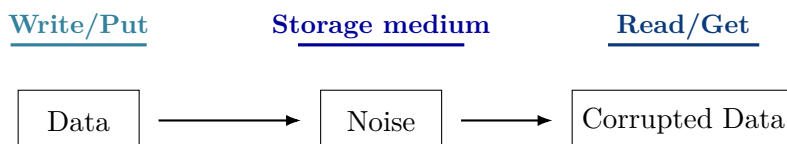
This reflects settings where multiple antennas transmit and receive simultaneously.

2.7 Storage systems

Coding theory is not only about sending data through space. It also applies to storing data over time:

write/put $\longrightarrow$ storage medium $\longrightarrow$ read/get.

Noise or faults in the storage medium can corrupt the retrieved information.



The same erasure and error ideas appear here as well. For storage systems, *transmission/reception* is replaced by *put/get* operations.

3 Coding Theory as a Subject

A compact description is:

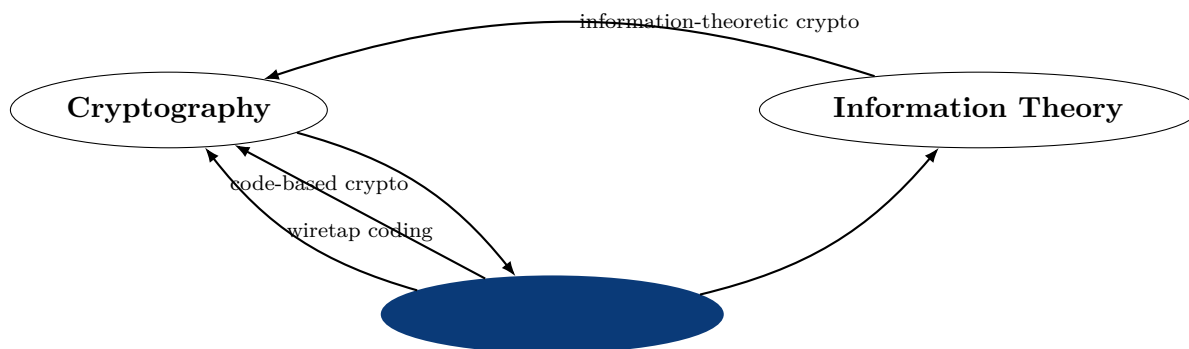
Working definition

Coding theory is the study of the design of mechanisms that ensure reliability of data transmission in the presence of perturbation. For storage systems, the same viewpoint applies with write/read operations in place of transmit/receive operations.

In many introductory courses, the emphasis is placed on transmission through discrete channels, because that setting already contains the central ideas of encoding, redundancy, and decoding.

4 Related Areas

Coding theory sits naturally between several nearby disciplines.



- **Information theory** studies fundamental limits on communication, such as capacity, compression, and noise tolerance.

- **Cryptography** studies secrecy, authenticity, and adversarial security.
- **Coding theory** overlaps with both: for example, code-based cryptography and wiretap coding lie directly at these interfaces.

5 Applications

Coding theory appears in many settings:

- coding for communications (wired and wireless),
- coding for storage (CDs, RAID systems, cloud storage),
- network coding,
- coding for memories,
- quantum coding,
- source coding and compression.

Why coding theory matters

The subject is not a narrow niche. It is a common mathematical language for reliability across communication, storage, networking, and even quantum information.

6 The Binary Erasure Channel

One of the simplest and most important toy models is the *binary erasure channel* (BEC).

6.1 Definition

The BEC has:

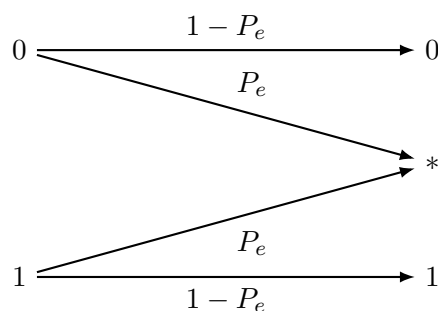
- input alphabet $\{0, 1\}$,
- output alphabet $\{0, 1, *\}$,
- erasure probability P_e .

If the input bit is 0, then the output is:

$$0 \xrightarrow{1-P_e} 0, \quad 0 \xrightarrow{P_e} *.$$

If the input bit is 1, then the output is:

$$1 \xrightarrow{1-P_e} 1, \quad 1 \xrightarrow{P_e} *.$$



6.2 Interpretation

The channel never flips 0 into 1 or 1 into 0. Instead, it either preserves the bit or replaces it by an erasure symbol $*$. The benefit is that the receiver knows exactly which coordinate was lost.

6.3 Example of a transmitted word

A binary input word such as

$$(0, 1, 0, 0, 1) \in \{0, 1\}^5$$

may be received as

$$(0, 1, 0, *, 1).$$

Only one coordinate was erased, and its position is visible.

7 Conceptual Comparison of Channel Types

Model	What happens?	What the receiver knows
Erasure channel	Some symbols are replaced by $*$ or marked missing	The uncertain locations are known
Error channel	Some symbols are changed into other valid symbols	The uncertain locations are usually unknown
Discrete channel	Symbols come from a finite alphabet A	Works with block codes over finite alphabets
Continuous channel	Symbols are real or complex valued	Noise is modeled in a continuous signal space
Storage model	Data degrades between write and read	Reliability is needed over time, not just distance

8 Why Redundancy Is Necessary

Suppose a sender transmits data with no redundancy at all. Then any erased or corrupted symbol may destroy information permanently. Coding theory answers this by replacing the

original message with a longer, more structured codeword.

Typical design principles include:

- adding redundant symbols,
- choosing codewords with enough separation,
- designing efficient decoding rules,
- exploiting special channel knowledge (for example, erasure locations).

This introductory lecture does not yet build specific codes, but it sets up the language used later when one studies Hamming distance, linear codes, parity checks, generator matrices, syndrome decoding, channel capacity, and more advanced constructions.

9 Short Summary

Main takeaways

1. Coding theory studies reliable communication and storage in the presence of noise.
2. The basic distinction is between *erasures* and *errors*.
3. Models may be discrete, continuous, matrix-valued, or storage-based.
4. The subject is closely related to information theory and cryptography.
5. The binary erasure channel is a fundamental first example.

Possible next topics

If you continue these notes into a full course, natural next sections would be:

- Hamming distance and nearest-neighbor decoding,
- block codes and linear codes,
- parity-check and generator matrices,
- erasure correction versus error correction,
- repetition codes, parity codes, and Hamming codes,
- channel capacity and Shannon's viewpoint.